Ch: 14 Partial derivatives

$f(x) = x + 1 \rightarrow y = x + 1$ 1 Independent Variable (x)

To graph this we have X-y Plane

Rule

To graph a function you must have one more dimension [(e.g axis)] then the total no. of **Independent** Variables.

$f(x, y)$ ☆ $\rightarrow$ no. of variables +1 = dimension

$2 + 1 =$ 3D (x, y, z)

↓

two Independent Variables

(In order to get an output you now need 2 numbers

You need a point (x, y)

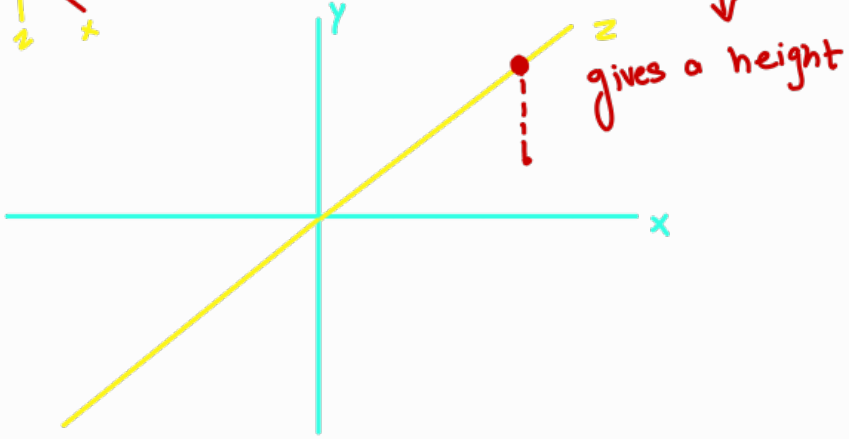
to get a Result

(a.k.a graph is a surface)



$g(x, y) = x^2 + y^2$ Same as

$z = x^2 + y^2$ (3-D)



$h(x,y,z) = \dfrac{x^2+y^2}{z-3}$

Dimension = 3 + 1 = 4D | Impossible

Number of points needed to generate a output is (x,y,z)

$$w = \frac{x^2+y^2}{z-3}$$

The max no. of dependent variables we can have is only 1.

To graph the domain of the function it has same no. of dimensions as that of no. of independent variables.

$$f(x,y,z) = \sqrt{x^2+2y^2+3z^2}$$

4-D graph

3-D Domain

$$f(0,2,-1) = \sqrt{0+2(2)^2+3(-1)^2}$$

$$= \sqrt{11}$$

Domain :- no. of Independent Variables

finding domain and Range

1) $f(x,y) = \frac{xy}{x-y}$

$x - y \neq 0$

$\boxed{x \neq y}$

Domain

$\boxed{x \neq y}$

(x,y)

↳ I can plug in anything except $\boxed{x \neq y}$ which is the domain

Range:/output

$z = \frac{xy}{x-y}$

$(-\infty, \infty)$

↳ will be for z only. / contains dependent variable.

2) $g(x,y) = \sqrt{4 - x^2 - y^2}$

Domain

$4 - x^2 - y^2 \geq 0$

$4 \geq x^2 + y^2$

$\boxed{x^2 + y^2 \leq 4}$

The graph of domain will be 2D circle

Range:

$z = \sqrt{4 - x^2 - y^2}$

$z = \sqrt{4 - (x^2 + y^2)}$

$\boxed{0 \leq z \leq 2}$

$z = \sqrt{4 - 5}$ X

$$\frac{\text{Anything}}{\sqrt{x^2+1}}$$ ☆

$x^2+1 \geq 0$ ✗

$x^2+1 > 0$ ✓

graphing domain

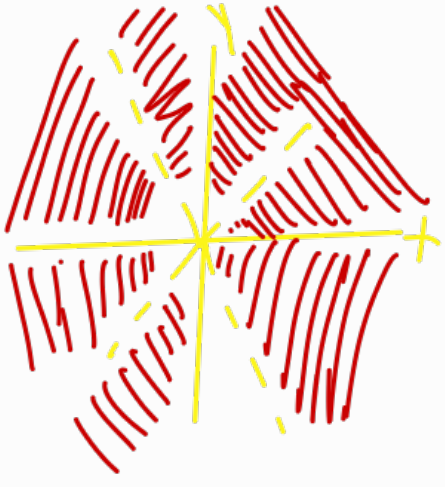
→ $f(x,y) = \frac{1}{\sqrt{x}}$ $x > 0$



→ $f(x,y) = \frac{xy}{x^2-y^2}$ $x^2 - y^2 \neq 0$

$x^2 \neq y^2$

$y \neq +x$ $y \neq -x$



± signs comes when you take a √ ★

→ $f(x,y) = \dfrac{\ln(y-x)}{\sqrt{x-y+1}}$

$x-y+1>0$

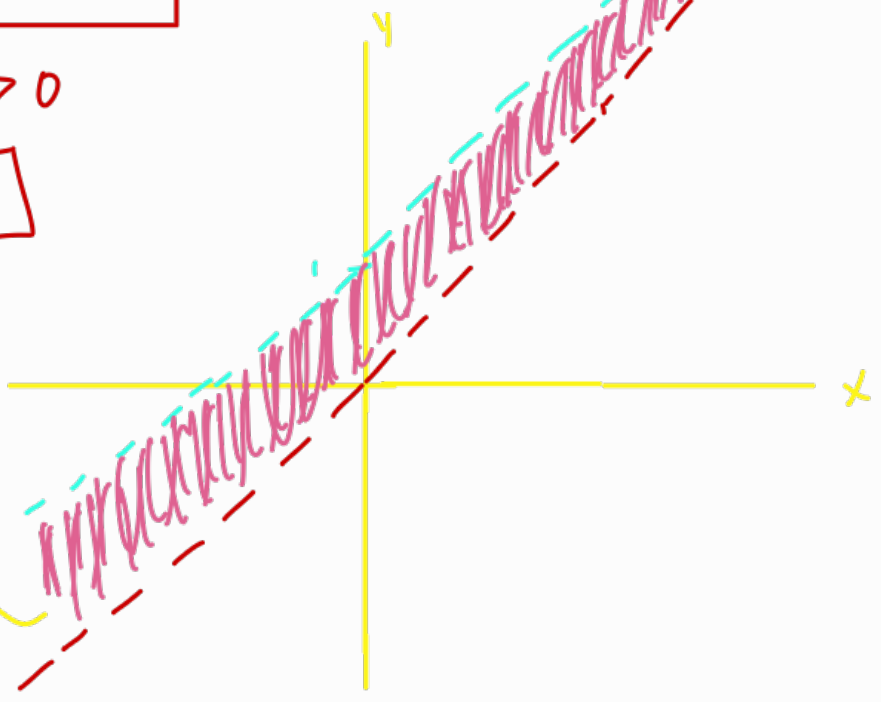
$x+1>y$

$y<x+1$

$\ln(y)$ $y>0$ e.g.

$y-x>0$

$y>x$



graphing $y=x$

dotted line as not equal or $\leq$ $\geq$

→ Shaded Region are the points where the actual graph of $f(x,y)$ will exist in 3D.

graphing domain 3 variables

(4-D graph 3-D Domain)

→ $f(x,y,z) = \sqrt{9 - x^2 - y^2 - z^2}$

Domain

$9 - x^2 - y^2 - z^2 \geq 0$

$9 \geq x^2 + y^2 + z^2$

$$\boxed{x^2 + y^2 + z^2 \leq 9}$$

↳ graph is a Sphere.

Range / output

$[0, 3]$?

→ $f(x,y,z) = \dfrac{\sqrt{4 - x^2 - y^2}}{z-3}$

Domain:

$z - 3 \neq 0$

$$\boxed{z \neq 3}$$

$4 - x^2 - y^2 \geq 0$

$4 \geq x^2 + y^2$

$$\boxed{x^2 + y^2 \leq 4}$$

↳ graph is a cylinder in 3D

	Domain	Range
$\sin(x)$	$(-\infty, \infty)$	$[-1, 1]$
$\cos(x)$	$(-\infty, \infty)$	$[-1, 1]$
$\tan(x)$	All Real numbers except odd multiples of 90°	$(-\infty, \infty)$

$= \frac{\sin\theta}{\cos\theta}$

$\cos\theta \neq 0$

$\theta \neq \cos^{-1}(0)$ $\left(\frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}\right)$

$\theta \neq 90°$ (90 odd multiples)

	Domain	Range
$\operatorname{cosec} x$	All real nos except $n\pi$, $n \in 1,2,3,4,5$	$(-\infty,-1] \cup [1,\infty)$
$\sec x$	All real no except 90° odd multiples	$(-\infty,-1] \cup [1,\infty)$
$\cot x$	All real no except $n\pi$	$(-\infty,\infty)$

(Disk made by vs)

Note
(Region is the domain)



Region

complete disk in Region

→ positive Radius

→ And Interior point is at centre of disk

Interior point only if



boundary point

A point that is at centre of a disk But the disk is not completely inside the Region R

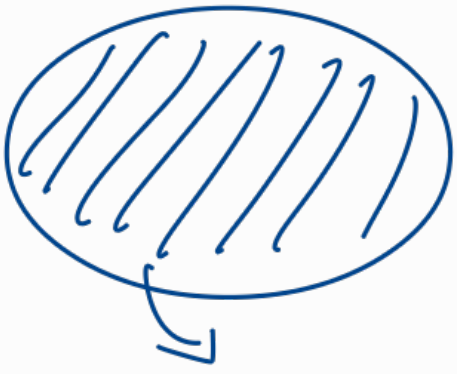
Interior point is included in Region but Boundary point may not be needed to be included in Region R

A region is set to be **closed** if all Boundary points and interior points included.

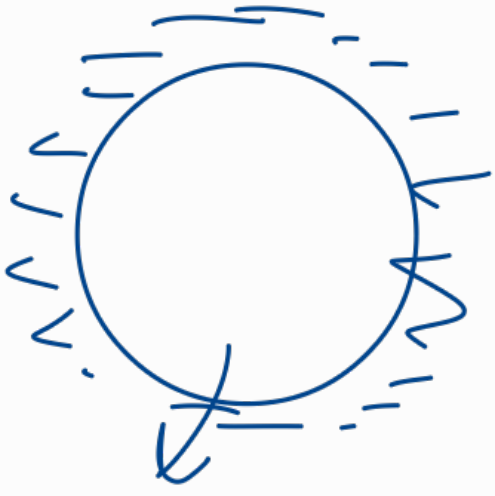
A region is set to be **open** if all interior points are included.

Note::

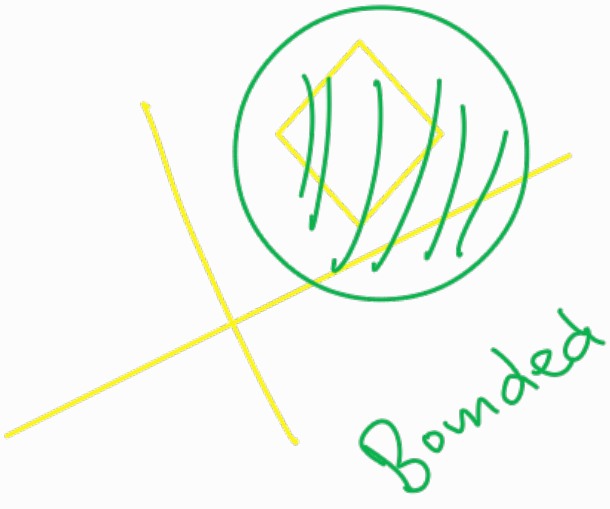
A region in a plane is bounded if it lies inside a disk of finite radius. Else it is unbounded.



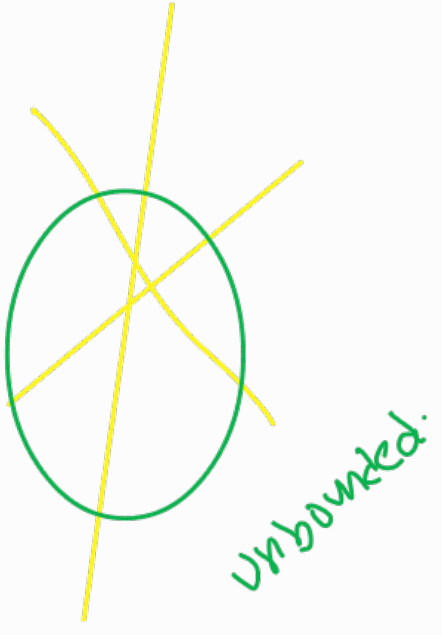
Region bounded



Unbounded



Bounded



Unbounded

Level curve

$f(x,y) = c$ ← fixed constant

going to be a curve in x-y plane (2-D)

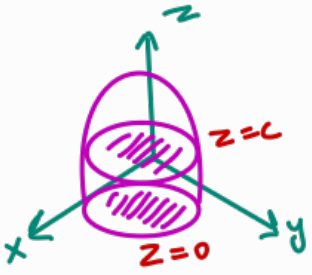
1

$f(x,y) = 100 - x^2 - y^2$

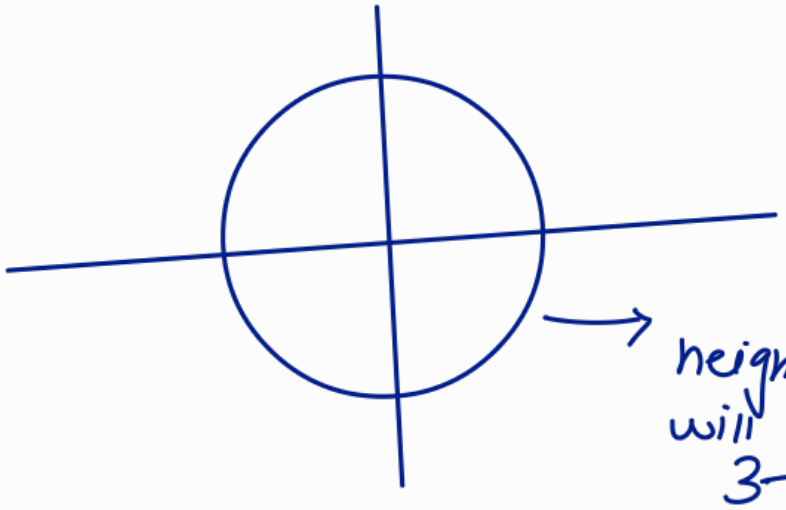
$c = 0$

$0 = 100 - x^2 - y^2$

$x^2 + y^2 = 100$



C is value of third axis



height will be in 3-D

find level curve:

$f(x,y) = \sqrt{x + y^2 - 3}$ $\quad (3,-1)$

$f(3,-1) = \sqrt{3 + 1 - 3}$

$= \sqrt{1} =$ $\boxed{1 = c}$

$\sqrt{x + y^2 - 3} = 1$

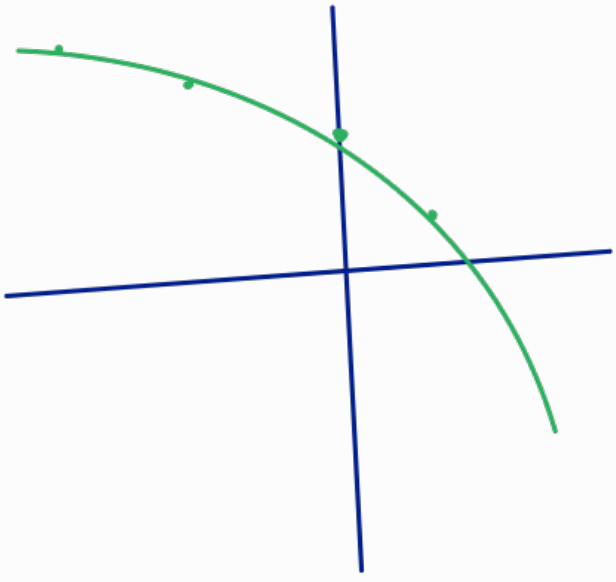
$x + y^2 - 3 = 1$

$y^2 + x - 3 - 1 = 0$

$y^2 + x - 4 = 0$

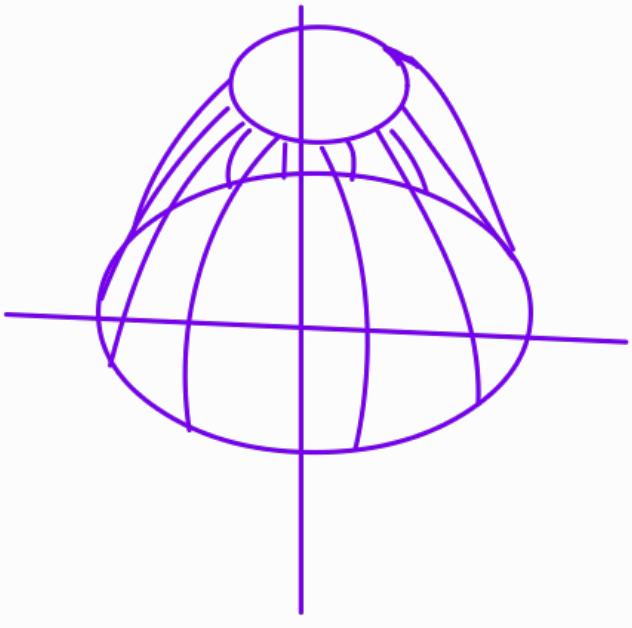
$y^2 = 4 - x$

$y = \sqrt{4 - x}$



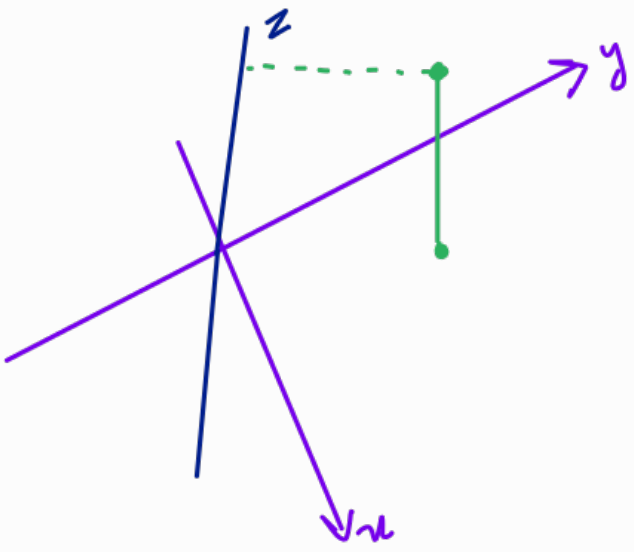
x	y
0	2
1	$\sqrt{3}$
-1	$\sqrt{5}$

Contour curve : A curve in Space (3-D) in which the plane $z = c$ cuts the curve in space.



Limits

$\lim_{x \to a} f(x) = L$ approaching a on the x-axis by travelling along the curve.



$\lim_{(x,y) \to (a,b)} f(x,y) = L$

↓

L is height / z-axis.

On a curve there are two ways you can approach | one variable

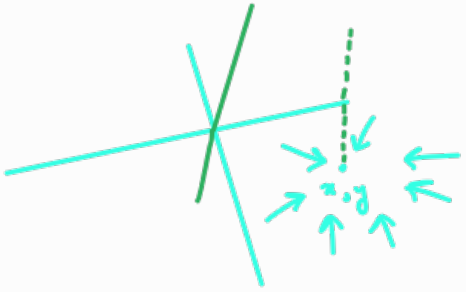
On a surface there are Infinite ways you can approach the limit | two variables.

To prove a limit exists we must prove that along all paths we approach the same point.



We can prove a limit DNE by showing that along 2 paths $f(x,y)$ we get a different value as we approach the same point (x,y).

$$\lim_{(x,y)\to(0,0)} \frac{x^2-y^2}{2x^2+y^2}$$



Along $x=0$ travelling along y-axis.

$$\lim_{y\to 0} \frac{0^2-y^2}{2(0)^2+y^2} = \frac{-y^2}{y^2} = \boxed{-1}$$

Along $y=0$

$$\lim_{x\to 0} \frac{x^2-0^2}{2x^2+0^2} = \boxed{\frac{1}{2}}$$ DNE

The limit DNE at $x=0$ $y=0$

---

1) Try $x=0$ and $y=0$

2) if this doesn't work chose another path that contains the point we are approaching

3) Substitute so Degrees of numerator and demonator become same

$$\lim_{(x,y)\to 0,0} \frac{3xy}{3x^2+y^2}$$

$x=0$

$$\frac{3(0)y}{3(0)^2+y^2} = \boxed{0}$$

$$\lim_{\substack{(x,y)\to 0,0 \\ y=0}} \frac{3(x)(0)}{3x^2+0^2} = \boxed{0}$$

Along $y=x$ → goes through $(0,0)$

$$\frac{3x^2}{3x^2+x^2} \underset{x\to 0}{=} \frac{3x^2}{4x^2} = \boxed{\frac{3}{4}}$$

DNE

$$\lim_{(x,y)\to(0,0)} \frac{xy^3\cos x}{2x^2+y^6}$$

$x=0$

$$\frac{0}{2(0)^2+y^6} = \boxed{0}$$

$y=0$

$$\frac{xy^3\cos x}{2x^2+y^6}$$

$$\frac{0}{2x^2} = \boxed{0}$$

now using $y=x$

$$\frac{x\cdot x^3\cos x}{2x^2+x^6} = \frac{x^4\cos x}{2x^2+x^6} = \frac{0}{0}$$

undefined

$\boxed{\text{DNE}}$

1) $f(x,y)$ along x-axis

2) $f(x,y)$ along y-axis

3) $f(x,y)$ along $y = mx$

4) $f(x,y)$ along $y = kx^2$

$$f(x,y) = \frac{x^4}{x^4+y^2}$$

$$\lim_{(x,y)\to(0,0)} \frac{x^4}{x^4+y^2}$$

Along $x = 0$

$$x \to 0 \quad \frac{(0)^4}{(0)^4+y^2} = \boxed{0}$$

Along $y = 0$

$$y \to 0 \quad \frac{x^4}{x^4} = \boxed{1}$$

DNE

$$f(x,y) = \frac{x^4 - y^2}{x^4 + y^2}$$

$$\lim_{(x,y) \to 0,0} \frac{x^4 - y^2}{x^4 + y^2}$$

Along $x = 0$

$$\frac{(0)^4 - y^2}{(0)^4 + y^2} = \boxed{-1}$$

Along $y = 0$

$$\frac{x^4 - (0)^2}{x^4 + (0)^2}$$

$\boxed{1}$

DNE

Partial derivatives

there are infinite no of tangents to a surface at a point.

$$f(x,y) = 7x^2 - x^3y^4 + 5x^4y^3$$

f_x (derivative WRT x) every other thing is constant. here y is kept constant

$$14x - 3x^2y^4 + 20x^3y^3$$

$$f_y = 7x^2 - x^3y^4 + 5x^4y^3$$

$$f_y = 7x^2 - x^3 4y^3 + 5x^4 3y^2$$

$$f_y = 0 - 4y^3x^3 + 3y^2 5x^4$$

$$Z = e^{x^2y^3}$$

find

$$\frac{dz}{dx} = e^{x^2y^3} \times 2xy^3$$

$$\frac{dz}{dy} = e^{x^2y^3} \times 3y^2x^2$$

$$f(x,y) = \ln(x^2 + y^2)$$

$$\ln u = \frac{u'}{u}$$

$$\frac{dz}{dx} = \frac{2x}{x^2+y^2}$$

$$\frac{dz}{dy} = \frac{2y}{x^2+y^2}$$

$$f(x,y) = x^3 e^{4xy^2}$$

we used product Rule only for $\frac{dz}{dx}$ as x was in both functions but don't need PR for $\frac{dz}{dy}$ as it is only in $e^{\square}$.

$\frac{dz}{dx}$ (using product Rule)

$$3x^2 e^{4xy^2} + x^3 e^{4xy^2} \times 4y^2$$

$$\frac{dz}{dy} = x^3 e^{4xy^2} \times 8xy$$

find Partial derivative for

$$\frac{u'v - v'u}{v^2}$$

$$f(x,y) = \frac{3y^5}{x^3+y^3}$$

$$\frac{dz}{dy} = \frac{15y^4(x^3+y^3) - (0+3y^2)(3y^5)}{(x^3+y^3)^2}$$

$$\frac{dz}{dy} = \frac{15y^4(x^3+y^3) - 9y^7}{(x^3+y^3)^2}$$

$$\frac{dz}{dy} = \frac{15x^3y^4 + 15y^7 - 9y^7}{(x^3+y^3)^2}$$

$$\frac{dz}{dy} = \frac{15x^3y^4 + 6y^7}{(x^3+y^3)^2}$$

$$\boxed{\frac{dz}{dy} = \frac{3y^4(5x^3+2y^3)}{(x^3+y^3)^2}}$$

$$\frac{dz}{dx} = 3y^5(x^3+y^3)^{-1}$$

$$= -3y^5(x^3+y^3)^{-2} \times 3x^2 + 0$$

$$= \frac{-9y^5x^2}{(x^3+y^3)^2}$$

$$f(x,y,z) = x^5y^2z^4$$

$$\frac{dh}{dz} = 4x^5y^2z^3$$

$$\frac{dh}{dx} = 5x^4y^2z^4$$

$$\frac{dh}{dy} = 2x^5yz^4$$

$$\boxed{(f\cdot g\cdot h)' = f'gh + fg'h + fgh'}$$

Three Variable P.R

find Second partial derivative

f_{xx}, f_{xy}, f_{yy}, f_{yx} -> first take derivative WRT y and then WRT

f_{xy} → first take derivative WRT x and WRT y

$f = x^3 + 4x^5y^3 + 5y^4$

$f_{xx} = 3x^2 + 20x^4y^3 + 0$

$f_{xx} = 6x + 80x^3y^3$

$f_{yx} = 0 + 12y^2x^5 + 20y^3$

$f_{yx} = 0 + 60y^2x^4$

$f_{xy} = 3x^2 + 20x^4y^3 + 0$

$f_{xy} = 0 + 20x3x^4y^2$

$f_{xy} = 60x^4y^2$

Chain Rule

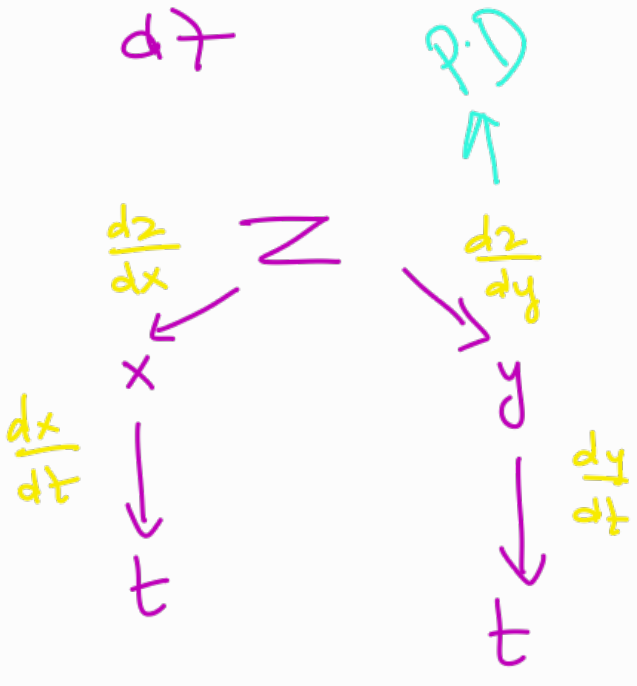
$z = x^3y + 3x^2y^3$

$x = 4 + t^2 \qquad y = 5t^3$

find $\frac{dz}{dt}$

z is a function of x and y

x and y are a function of t.



$$\frac{dz}{dt} = \frac{dz}{dx} \times \frac{dx}{dt} + \frac{dz}{dy} \times \frac{dy}{dt}$$

$z = x^3y + 3x^2y^3$

$\frac{dz}{dx} = 3x^2y + 6xy^3$

$\frac{dz}{dy} = x^3 + 3y^2 3x^2$

$\frac{dz}{dy} = x^3 + 9y^2x^2$

$x = 4 + t^2 \qquad y = 5t^3$

$\frac{dx}{dt} = 0 + 2t \qquad \frac{dy}{dt} = 15t^2$

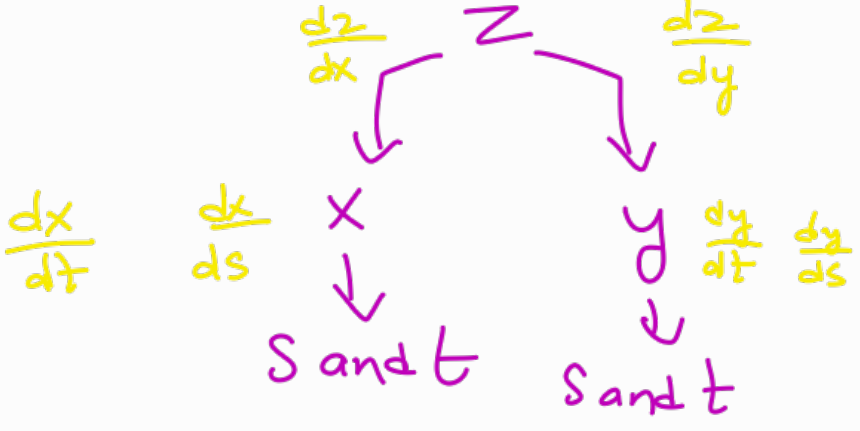
$2t \times (3x^2y + 6xy^3) + 15t^2(x^3 + 9y^2x^2)$

(Next problem)

$z = 4x^3y^5 \qquad x = 3s^2t \qquad y = 2s^3t^2$

find

$\frac{dz}{ds}$ and $\frac{dz}{dt}$



finding:

$\frac{dz}{ds} = \frac{dz}{dx} \times \frac{dx}{ds} + \frac{dz}{dy} \times \frac{dy}{ds}$

$$\frac{dz}{dt} = \frac{dz}{dx} \times \frac{dx}{dt} + \frac{dz}{dy} + \frac{dy}{dt}$$

$z = 4x^3y^5 \quad x = 3s^2t \quad y = 2s^3t^2$

$\frac{dz}{dx} = 12x^2y^5 \qquad \frac{dx}{ds} = 6st \qquad \frac{dy}{ds} = 6s^2t^2$

$\frac{dz}{dy} = 20y^4x^3 \qquad \frac{dx}{dt} = 3s^2 \qquad \frac{dy}{dt} = 4ts^3$

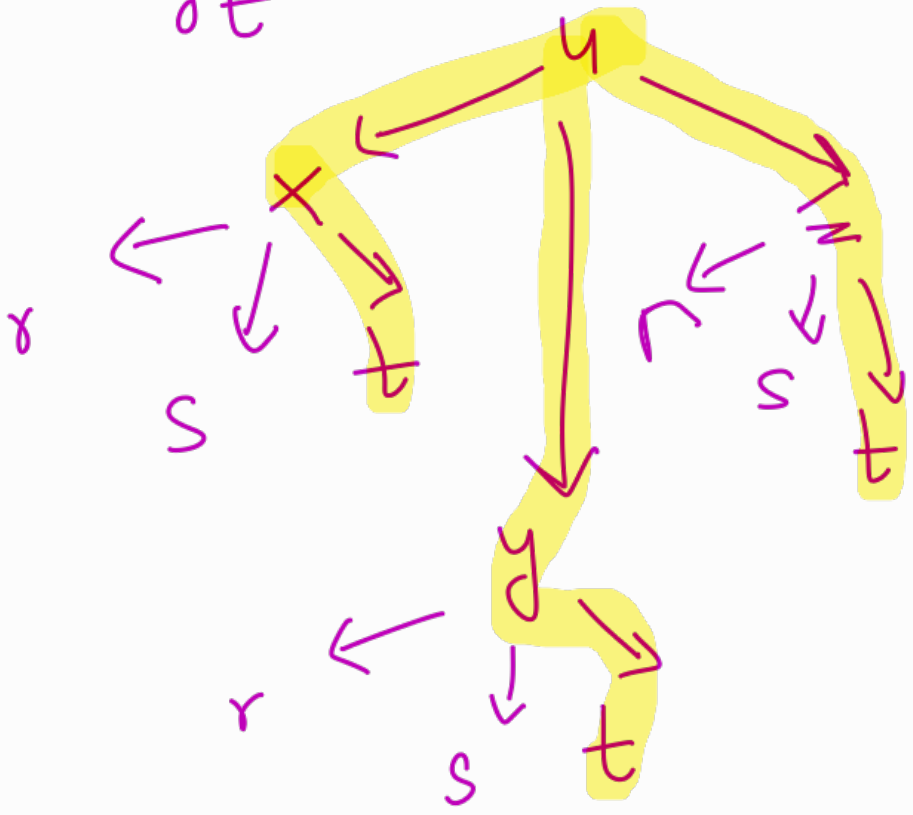
$$\frac{dz}{ds} = (12x^2y^5 \times 6st) + (20y^4x^3 \times 6s^2t^2)$$

$$\frac{dz}{dt} = (12x^2y^5 \times 3s^2) + (20y^4x^3 \times 4ts^3)$$

$u = x^2y + y^3z^2 \quad x = rs^2t \quad y = r^2st^3$

$z = rst$

find $\frac{du}{dt}$



$$\left(\frac{du}{dx} \times \frac{dx}{dt}\right) + \left(\frac{du}{dy} \times \frac{dy}{dt}\right) + \left(\frac{du}{dz} \times \frac{dz}{dt}\right)$$

$u = x^2y + y^3z^2 \quad x = rs^2t \quad y = r^2st^3$

$z = rst$

$\frac{dy}{dx} = 2xy + 0 \times \frac{dx}{dt} = rs^2$

$\frac{dy}{dy} = x^2 + 3y^2z^2 \times \frac{dy}{dt} = 3t^2r^2s$

$\frac{dy}{dz} = 2z^2y^3 \times \frac{dz}{dt} = rs$

$(2xy) \times (rs^2) + (2z^2y^3 \times rs) + (x^2 + 3y^2z^2 \times 3t^2r^2s)$

(Implict derivatives) equations in form $f(x,y)=0$

$$\frac{dy}{dx} = \frac{-F_x}{F_y}$$ to use this eq must be $=0$

e.g find $\frac{dy}{dx}$ for $y^2 - x^2 - \sin xy = 0$

$\frac{-F_x}{F_y}$ ↓ (partial)

$$\frac{df}{dx} = 0 - 2x - \cos xy \times y = 0$$

$$\boxed{\frac{df}{dx} = -2x - y\cos xy}$$

$$\boxed{\frac{df}{dy} = 2y - \cos xy \times x}$$

$$\frac{dy}{dx} = -\left(\frac{-2x - y\cos xy}{2y - x\cos xy}\right) = -\left(\frac{-2x-y}{2y-x}\right)$$

$$= \boxed{\frac{2x+y}{2y-x}}$$

note

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} \quad \text{and} \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z}$$

↓ partial derivative

find

$\frac{dz}{dx}$?

$$x^3 + z^2 + ye^{xz} + z\cos y = 0$$

$$F_x = 3x^2 + 0 + ye^{xz} \cdot xz + 0 = 0$$

$$\boxed{F_x = 3x^2 + zye^{xz}}$$

$$F_y = 0 + 0 + e^{xz} + z(-\sin y)$$

$$\boxed{F_y = e^{xz} - z\sin y}$$

$$F_z = 0 + \partial z + ye^x x + \cos y$$

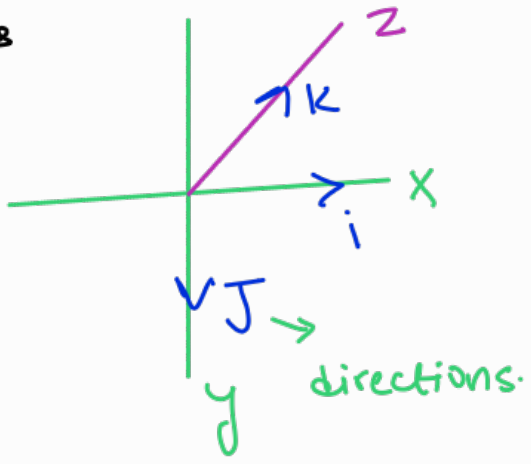
$$\boxed{F_z = \partial z + xye^x + \cos y}$$

$$\frac{dz}{dx} = -\frac{F_x}{F_z}$$

$$\frac{-3x^2 - zye^{xz}}{\partial z + xye^x + \cos y}$$

Directional derivatives

finding derivatives in any direc.



$\frac{df}{dx}$ = slope of tangent line in direction of i.

To check variation of slope in any other direction than i and j we use direction derivative.

$$\left(\frac{df}{ds}\right)_{\hat{u}, P_0} = \lim_{s \to 0} \frac{f(x_0 + su_1, y_0 + su_2) - f(x_0, y_0)}{s}$$

- $\hat{u}$: unit vector gives Direction
- P_0: point
- $s \to 0$: distance b/w two points
- ↓ Definition of Direction derivative

Notation

$$\left(\frac{df}{ds}\right)_{\hat{u}, P_0}$$

$$\hat{u} = \frac{\vec{u}}{|\vec{u}|}$$

if $\hat{u} = \hat{i}$ then we find partial derivative of x. $\frac{\partial f}{\partial x}$ and if $\hat{u} = \hat{j}$ then partial derivative is of y. $\frac{\partial f}{\partial y}$

e.g $\vec{u} = 4i + 3j$ Find change in f at $P_0(5,5)$ when moving along $f(x,y) = 2xy - 3y^2$

$$\hat{u} = \frac{\vec{u}}{|\vec{u}|} = \frac{4i+3j}{\sqrt{(4)^2+(3)^2}} = \frac{4i+3j}{5} = \frac{4i}{5} + \frac{3j}{5}$$

$$\left(\frac{df}{ds}\right)_{\hat{u}, P_0} = \lim_{s\to 0} \frac{f(x_0 + su_1, y_0 + su_2) - f(x_0, y_0)}{s}$$

$$\lim_{s\to 0} \frac{f\left(5 + \frac{4}{5}s, 5 + \frac{3}{5}s\right) - f(5,5)}{s}$$

$f(5,5) = -25$

putting $f\left(5 + \frac{4}{5}s, 5 + \frac{3}{5}s\right)$ in equation.

$$f(x,y) = 2\left(5 + \frac{4}{5}s\right)\left(5 + \frac{3}{5}s\right) - 3\left(5 + \frac{3}{5}s\right)^2$$

$$= 2\left(\frac{25+4s}{5}\right)\left(\frac{25+3s}{5}\right) - 3\left(\frac{25+3s}{5}\right)^2$$

$$= -25 - 4s - \frac{3}{25}s^2$$ After Simplification.

$$\lim_{s\to 0} \frac{\left[-25 - 4s - \frac{3}{25}s^2 - (-25)\right]}{s} = \boxed{-4}$$

gradient of function

(Ans to the compute $\frac{df}{ds}$)

$$\nabla f = \frac{\partial f}{\partial x} i + \frac{\partial f}{\partial y} j$$

is in (x-y plane)

$$\frac{df}{ds} = \left| \nabla f(P_0) \cdot \hat{u} \right| = \nabla f \cos\theta$$

- $\frac{df}{ds}$: Directional Derivative
- P_0 : Specific point
- $\cdot$: Dot product

Properties

1) f increases most rapidly (Max) when $\theta = 0$

u is in direction of gradient.

2) f decreases most when $\theta = \pi$.



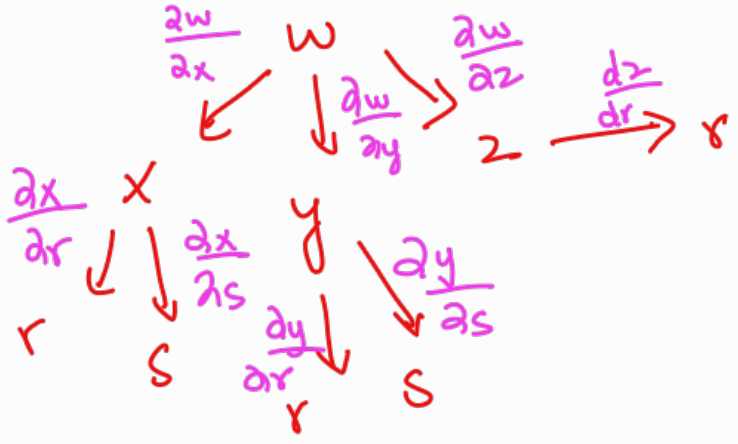
3) f remains Same if $\theta = \frac{\pi}{2}$.



(zero change)

note $i \cdot i = 1$
$j \cdot j = 1$
$k \cdot k = 1$
} Dot product

$w = x + 2y + z^2 \qquad x = \frac{r}{s} \qquad y = r^2 + \ln s \qquad z = 2r$



$\frac{\partial w}{\partial r}$ and $\frac{\partial w}{\partial s}$

$$\frac{\partial w}{\partial s} = \left(\frac{\partial w}{\partial x} \cdot \frac{\partial x}{\partial s}\right) + \left(\frac{\partial w}{\partial y} \cdot \frac{\partial y}{\partial s}\right)$$

$$\frac{\partial w}{\partial x} = 1 \qquad \frac{\partial w}{\partial y} = 2$$

$$\frac{\partial x}{\partial s} = rs^{-1} = -rs^{-2} \qquad \frac{\partial y}{\partial s} = \frac{1}{s}$$

$$1 \times (-rs^{-2}) + \left(2 \times \frac{1}{s}\right)$$

$$-\frac{r}{s^2} + \frac{2}{s}$$

$$\boxed{\frac{2}{s} - \frac{r}{r^2}}$$

(note:)

$$\frac{df}{ds} = \nabla f \cdot \hat{u} = |\nabla f| \quad \text{(mag)}$$

increases most Rapidly

$$\frac{df}{ds} = \nabla f \cdot \hat{u} = -|\nabla f|$$

decreases most Rapidly

e.g $f(x,y,z) = \frac{x}{y} - yz$ $\quad P_0(4,1,1)$

find the max and min values.

$$\nabla f = \frac{\partial f}{\partial x} i + \frac{\partial f}{\partial y} j + \frac{\partial f}{\partial z} k$$

$$f(x,y,z) = \frac{x}{y} - yz$$

$$\frac{\partial f}{\partial x} = \boxed{\frac{1}{y}} \qquad \frac{\partial f}{\partial y} = \boxed{-xy^{-2} - z}$$

$$\frac{\partial f}{\partial z} = 0 - y = \boxed{-y}$$

$$1i - 5j - 1k \Rightarrow \boxed{i - 5j - 1k}$$

Derivative of f is max in direction of $i - 5j - k$ and has min value in direction of $-(i - 5j - k)$ and their magnitudes are $|\nabla f|$ and $-|\nabla f|$

for largest and smallest use ∇f.

$$\vec{u} = \frac{\nabla f}{|\nabla f|} \rightarrow \text{largest}$$

$$\vec{u} = -\frac{\nabla f}{|\nabla f|} \rightarrow \text{smallest}$$

this gives direction

Ex 14.5

Q 31 In what direction is the derivative of $f(x,y) = xy + y^2$ at $P(3,2)$ equal to zero.

$$\nabla f = f_x \hat{i} + f_y \hat{j} = y\hat{i} + (x+2y)\hat{j}$$

$$\nabla f(3,2) = 2\hat{i} + 7\hat{j}$$

A vector orthogonal or 90° is $2\hat{i} - 7\hat{j}$

$$\hat{u} = \frac{\vec{u}}{|\vec{u}|} = \frac{7}{\sqrt{53}}\hat{i} - \frac{2}{\sqrt{53}}\hat{j}$$

Another direction

$$-\hat{u} = \frac{-\vec{u}}{|\vec{u}|} = -\frac{7}{\sqrt{53}}i + \frac{2}{\sqrt{53}}j$$

$\nabla f \cdot \hat{u} = 0$ if they are $\perp$ 90°

Basically dot product of $2i + 7j$ with $2i - 7j$ gives 0 Ans

Q28 let $f(x,y) = x^2 - xy + y^2 - y$. find the directions u and the values of $D_u f(1,-1)$ for which $D_u f(-1,1)$ is largest, smallest, $D_u f(1,-1) = 0$ and y

(a) $\nabla f = 3\vec{i} - 4\vec{j} \Rightarrow |\nabla f| = 5$

Thus $\vec{u} = \dfrac{\nabla f}{|\nabla f|} = \dfrac{3i}{5} - \dfrac{4}{5}j$

↓ largest

Smallest $-\dfrac{3i}{5} + \dfrac{4}{5}j$

90% data created in 2 years alone

→ digital Businesses

→ Trad (Brick or mortar)

↳ no omni presence

→ Omni - channel

Brick or click / omni

Cloud → based platforms ↑ ramp up

Computational capacity

open source

revolution

Software code is freely avail

Eg github